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Fachbereich Humanwissenschaften
Institute of Cognitive Science

Bachelor's Thesis

Measuring Drift in Therapeutic AI: A Stability-Based Evaluation of Sovereign LLMs in Nightmare Therapy

Daniel Menzel
Bachelor's Program Cognitive Science

First supervisor: Moritz Hartstang
Institute of Cognitive Science
Osnabrück

Second supervisor: Prof. Dr. Sebastian Musslick
Institute of Cognitive Science
Osnabrück

Declaration of Authorship

The content of this thesis represents my own knowledge, my own understanding and my own perspective on the topic. In case artificial intelligence tools were used, their way and purpose of usage has been made transparent. Moreover, I have cited all my sources in accordance with academic standards. I am ready and able to explain and defend the positions developed in this thesis. This thesis has not been submitted, either in part or whole, at this or any other university.

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Abstract

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1. Introduction

1.1. Nightmare Disorder and Imagery Rehearsal Therapy

Nightmare disorder affects an estimated 2–5% of the general adult population (Li et al., 2010; Sandman et al., 2013; Schredl, 2010) and frequently co-occurs with PTSD and depression (Morgenthaler et al., 2018). The recommended treatment, Imagery Rehearsal Therapy (IRT), is a structured five-stage protocol in which the patient rewrites a recurring nightmare under therapist guidance (Krakow & Zadra, 2006). A persistent shortage of trained therapists limits the availability of IRT (Wittchen et al., 2011), making scalable, AI-assisted delivery a practical goal. However, the rewriting stage requires active strategy decisions from the therapist, and an AI agent that selects different strategies on repeated runs of the same case is clinically unreliable regardless of how capable any single response may be.

1.2. AI-Assisted Psychotherapy

Large language models are increasingly deployed in mental health, from empathy-based conversational tools to protocol-driven agents that deliver structured interventions such as cognitive behavioural therapy (Stade et al., 2024; Q. Zhang et al., 2025). reDreamAI is the deployment context for this study: a consumer-facing IRT chatbot that guides users through all five protocol stages. It is not a research prototype but a real deployment target where reliability is non-negotiable.

This deployment context creates a specific problem. A general-purpose assistant has no therapeutic obligation; response variability is acceptable and even desirable. A protocol-driven agent, by contrast, must make consistent strategic decisions. If the model selects different rescripting strategies for the same patient on different runs, the patient receives inconsistent care. Unlike rule-based chatbots whose responses can be reviewed once before deployment (Fitzpatrick et al., 2017), an LLM agent generates a new response on every run. Each run is a new draw from the output distribution, so stability must be evaluated repeatedly rather than verified once.

IRT targets nightmares rather than psychiatric diagnoses, which lowers the clinical risk relative to full psychotherapy. Ethical considerations are discussed in section 6.5 and section 7.2. The EU AI Act classifies autonomous therapeutic AI as high-risk and requires auditability (Aboy et al., 2024; van Kolfschooten & van Oirschot, 2024). Data sovereignty requirements push toward EU-resident models (Dale, 2025), introducing a constraint on model selection discussed in section 3.8.

1.3. The Evaluation Gap

Standard NLP benchmarks measure what a model *can* do, not whether it does the same thing reliably. A model can score well on a single evaluation while producing different therapeutic strategies on repeated runs of the same input. Capability and stability are distinct properties, but current benchmarks conflate them.

The closest existing work is BOLT (Chiu et al., 2024), which evaluates LLM therapist behaviour using 13 behavioural codes from motivational interviewing. BOLT measures whether a model applies appropriate therapeutic techniques (quality). This study measures a different property: whether a model applies the *same* techniques across repeated runs (stability). BOLT evaluates single responses; this study evaluates the distribution of responses across twenty independent trials per condition.

Other therapeutic AI evaluations are qualitative, post-hoc, or single-session (Talat et al., 2024). They cannot detect cross-run inconsistency because they do not present the same input multiple times. Systematic, reproducible in silico trials offer a way to measure stability at scale without requiring human participants.

1.4. Research Objectives

This thesis addresses the following research question:

> How consistent are LLM therapeutic responses across independent runs when given the same patient context within a structured IRT protocol?

Two contributions follow from this question.

Contribution 1. A three-level evaluation framework that measures stability at three layers: *plan consistency* (do strategy selections repeat?), *response similarity* (do the generated texts convey the same meaning?), and *plan-response alignment*

(does the model do what it declared?). Each layer captures a different aspect of therapeutic stability, and their combination reveals divergence patterns that no single metric can detect.

Contribution 2. A cross-model comparison that applies this framework to sovereign, proprietary, and open-weight models under controlled temperature conditions.

The study isolates the IRT rescripting phase because it is the stage that requires active strategy decisions. Isolating a single stage eliminates stage-routing confounds and allows any measured instability to be attributed to the therapist model rather than to upstream variation in the conversation. chapter 3 describes the experimental design in detail.

2. Background

2.1. Imagery Rehearsal Therapy

Nightmare disorder affects an estimated 2–5% of the adult population (Li et al., 2010; Sandman et al., 2013; Schredl, 2010) and shows strong comorbidity with Post-Traumatic Stress Disorder (PTSD) and depression (Morgenthaler et al., 2018). For patients with recurrent nightmares, the distress extends beyond the dream itself: sleep avoidance, daytime fatigue, and heightened anxiety create a cycle that standard sleep hygiene cannot break. IRT is the recommended treatment for this condition (Krakow & Zadra, 2006; Morgenthaler et al., 2018).

IRT follows a structured five-stage protocol. In the *recording* stage, the patient describes the nightmare in detail. In the *rewriting* stage, the patient and therapist collaboratively transform the nightmare narrative to reduce distress and increase a sense of mastery. The remaining stages (*summary*, *rehearsal*, and *final*) consolidate the new narrative and prepare the patient to rehearse it before sleep. Krakow and Zadra (2006) describe the rewriting instruction as deliberately open-ended: patients are told to "change the nightmare any way you wish."

The rewriting stage is the cognitive core of the protocol. Unlike recording (which requires listening) or rehearsal (which requires repetition), rewriting demands active therapeutic strategy decisions. The therapist must guide the patient toward a specific rescripting mechanism: confronting the threat, introducing protective elements, reframing the narrative, or transforming the dreamer's role within it. These choices are not arbitrary. Clinical research has identified distinct rescripting strategies with different prevalence rates in successful treatment. Harb et al. (2012) coded five categories from combat veteran rescripts: alternative endings (58%), positive image insertion (23%), threat transformation (13%), reality markers (10%), and distancing (8%). Germain et al. (2004) proposed a complementary framework through the Multidimensional Mastery Scale, which classifies rescripting outcomes along physical, social, environmental, emotional, mystical, and avoidance dimensions.

Treatment fidelity instruments provide the clinical standard for measuring whether a therapist delivers an intervention correctly. The IRT literature draws on two established frameworks: the ENACT rating scale (Kohrt et al., 2015) and the NIH BCC recommendations (Bellg et al., 2004). Both grade therapist adherence on a ternary scale: *absent*, *partial*, or *implemented*. This gradation captures clinically meaningful differences. A therapist who briefly mentions a rescripting strategy without developing it (partial) delivers a different intervention from one who fully guides the patient through it (implemented).

This ternary framework provides a scoring foundation that transfers directly to computational evaluation (see subsection 3.7.3). However, existing fidelity instruments assume a human clinician. They have not been adapted for Large Language Model (LLM)-based agents, where the therapist is a probabilistic text generator rather than a trained professional.

2.2. Treatment Fidelity in LLM-Based Interventions

The deployment of LLMs in mental health has expanded from empathy tools and symptom screeners to protocol-driven therapeutic agents. Stade et al. (2024) survey this trajectory and argue that LLMs could fundamentally reshape behavioural healthcare delivery, from diagnostic support to full psychotherapeutic interventions. Early systems such as Woebot (Fitzpatrick et al., 2017) delivered Cognitive Behavioral Therapy (CBT) through rule-based dialogue trees where every response was pre-authored and reviewed before deployment. For these systems, treatment fidelity is a design property: the chatbot cannot deviate from its script.

LLM-based agents operate differently. Each response is a fresh sample from the model's output distribution. A single model can produce high-quality, protocol-adherent responses on one run and strategically inconsistent ones on the next. This distinction is central to the treatment fidelity literature. Mowbray et al. (2003) formalise fidelity into two components: *adherence* (doing the prescribed things) and *competence* (doing them well). For rule-based chatbots, adherence is built in by construction. For LLM-based protocol agents, adherence becomes probabilistic and must be measured across runs rather than assumed from a single review.

LLM-based agents also face general failure modes such as hallucination, stage confusion, and persona inconsistency. These are well-documented concerns for clinical AI but fall outside the scope of this study. The present work focuses on a narrower problem: the probabilistic nature of LLM outputs creates drift even in

identical therapeutic contexts. When the same patient presents the same nightmare, stochastic sampling can produce different strategy selections, different therapeutic responses, and mismatches between declared and executed strategies. Each of these failure modes corresponds to a distinct layer of the generation process and requires a targeted evaluation method. Variable strategy selection maps to plan-level measurement (subsection 3.7.1), variable response content to semantic comparison (subsection 3.7.2), and plan-response mismatch to alignment verification (subsection 3.7.3).

Regulatory frameworks reinforce the need for stability measurement. The European Union (EU) AI Act classifies autonomous therapeutic AI as high-risk, imposing requirements for auditability, documentation, and risk management (Aboy et al., 2024; van Kolfschooten & van Oirschot, 2024). The Medical Device Regulation adds further constraints on software used in clinical contexts. For EU-based deployment scenarios, data sovereignty requirements push toward self-hosted, EU-resident models (Dale, 2025), constraining the model selection space (see section 3.8).

Despite these requirements, stability remains largely unmeasured. Prior therapeutic AI evaluation is qualitative, post-hoc, or single-session. Chiu et al. (2024) represent the closest prior work with their BOLT framework, which assesses LLM therapist behaviour across 13 behavioural codes drawn from motivational interviewing. BOLT asks whether the model does the right thing. The present study asks a different question: whether the model does the *same* thing across repeated runs. Existing frameworks, including BOLT, evaluate single responses rather than the distribution of responses across repeated trials.

2.3. Stochastic Behaviour in Language Models

At temperatures above zero, LLM token selection is probabilistic. The softmax distribution over the vocabulary is divided by the temperature parameter before sampling, so higher temperatures flatten the distribution and lower temperatures sharpen it toward the most likely token. At $T = 0$, the model selects the highest-probability token deterministically (greedy decoding). At $T > 0$, tokens are sampled from the resulting distribution, introducing variance into the output.

This per-token variance compounds through autoregressive generation. Each token is conditioned on all preceding tokens, so a single divergent token early in a response can cascade into a completely different therapeutic strategy. A model that selects “confront” rather than “imagine” at one token position may generate an en-

tirely different rescripting approach in the subsequent response. This *autoregressive cascading* means that the clinical impact of stochastic sampling is not proportional to the per-token entropy but can amplify through the response.

The clinical consequence is concrete: the same patient presenting the same nightmare on two different days may receive strategically inconsistent care. In a general-purpose assistant, this variability is benign or even desirable. In a manualized therapeutic protocol, where the therapist is expected to deliver consistent, evidence-based interventions, it represents a potential clinical failure.

Prior work on LLM stability addresses different questions than therapeutic consistency. Factual consistency studies measure whether models produce the same answers to knowledge questions across runs. Output diversity research quantifies lexical and semantic variation in creative generation. Benchmark reproducibility work examines whether evaluation scores are stable across random seeds. These lines of research share the concern for cross-run variation but do not address whether a model makes the same *clinical decisions* when presented with identical patient contexts.

The mechanical source of instability is well understood. What remains unquantified is its impact on structured therapeutic decision-making. How much does temperature-driven variance cost in clinical consistency? This question motivates the temperature sweep in the experimental design (section 3.9), which tests five points from $T = 0.0$ (deterministic) to $T = 0.6$ (typical vendor default for open-weight models).

2.4. Evaluation Metrics for Text Generation

Measuring consistency between text outputs requires a similarity metric. The choice of metric determines what kind of consistency is captured, and standard options fail in different ways for therapeutic language.

Surface-level metrics such as BLEU and ROUGE measure token overlap between texts. In therapeutic contexts, these metrics produce misleading results in both directions. Two responses that use identical words to describe different strategies score high on surface overlap despite being clinically divergent. Conversely, two responses that implement the same strategy through different phrasing score low despite being therapeutically equivalent. Therapeutic paraphrasing is the norm rather than the exception: a competent therapist will naturally vary their language

while maintaining the same strategic intent. Surface metrics penalise this valid variation.

BERTScore (T. Zhang et al., 2020) addresses this limitation by computing similarity in contextual embedding space rather than token space. Each token in the candidate text is matched to the most similar token in the reference text using cosine similarity over contextualised embeddings, producing precision, recall, and F1 scores. The metric captures semantic equivalence independent of surface form: two sentences that mean the same thing but use different words receive high scores.

The choice of embedding model affects BERTScore's behaviour. Among more than 130 available models, those fine-tuned on Natural Language Inference (NLI) tasks achieve the highest correlation with human judgements of semantic equivalence. DeBERTa-XLarge-MNLI (He et al., 2021) achieves the highest WMT16 human correlation ($r = 0.778$) and is selected for Method 2 in this study (subsection 3.7.2). Its disentangled attention mechanism handles the surface-form variation typical of therapeutic language, where valid paraphrasing is expected.

BERTScore captures semantic equivalence but cannot assess strategic intent. A metric that reports high similarity between two responses does not reveal whether both responses implement the same therapeutic strategy. The LLM-as-judge paradigm addresses this gap. By instructing a separate LLM to evaluate whether a response implements a declared strategy, it becomes possible to assess criteria that no scalar metric can capture: "does this response implement a confrontation strategy?" requires understanding both the strategy definition and the response content.

LLM-as-judge approaches carry known biases. Self-preference bias causes models to rate outputs from their own family more favourably. Position bias leads to higher scores for responses presented first. Verbosity bias rewards longer responses regardless of quality (Zheng et al., 2023). These biases can be mitigated through cross-model judging (the judge evaluates a different model family), structured rubrics, and explicit justification requirements. The mitigation strategy adopted in this study is detailed in subsection 3.7.3.

No single metric covers what clinical stability requires. Embedding similarity captures semantic equivalence but misses strategic intent. LLM judges can assess intent but introduce their own reliability concerns. Surface metrics capture neither. Existing therapeutic AI evaluations rely on a single metric type, leaving the gap between semantic equivalence and strategic intent unaddressed. This motivates the three-method design presented in chapter 3.

3. Methods

3.1. System Overview and Design Rationale

The evaluation system separates dialogue creation from stability measurement through two decoupled stacks. The *generation stack* produces synthetic therapy sessions by running a multi-agent dialogue loop. The *evaluation stack* measures the stability of therapeutic responses by presenting frozen conversation histories to the model under test and collecting multiple independent outputs.

This separation is a deliberate design choice. If the evaluation stack generated its own dialogue context, any measured instability could reflect upstream randomness in the patient or routing agents rather than genuine variation in the therapist model. By freezing the conversation history before evaluation begins, the system ensures that every trial receives identical input. Any observed inconsistency is attributable solely to the model under test.

The end-to-end data flow proceeds in five stages: patient vignettes feed the generation stack, which produces complete therapy sessions. These sessions are sliced at rewriting-turn boundaries to create frozen histories. The evaluation stack presents each frozen history to the therapist model across multiple independent trials, collecting both a strategy declaration and a therapeutic response per trial. Finally, an aggregation pipeline aligns all metrics across conditions for comparative analysis.

3.2. Dialogue Generation

The generation stack orchestrates a three-agent loop that produces naturalistic IRT sessions. Each turn proceeds through three agents in sequence:

1. The **patient agent** simulates a clinical presentation based on a Brief Dreamer Interview (BDI)-profiled vignette. Each vignette defines the patient's nightmare content, emotional presentation style, and engagement pattern.

2. The **router agent** classifies the current IRT stage from the conversation history, determining whether the session is in the recording, rewriting, summary, rehearsal, or final phase.

3. The **therapist agent** generates a stage-appropriate therapeutic response, guided by the router's classification and a stage-specific system prompt.

This three-agent architecture separates clinical reasoning from stage classification and patient simulation, matching the deployment architecture of reDreamAI. The router's stage classification controls which system prompt the therapist receives, ensuring that responses are appropriate to the current protocol phase.

Six patient vignettes cover the range of clinical presentation difficulty expected in IRT deployment: *anxious*, *avoidant*, *cooperative*, *resistant*, *skeptic*, and *trauma*. These profiles range from a cooperative patient who engages readily with rescripting instructions to a trauma patient whose nightmare content triggers intense emotional responses during the session. The vignette design draws on prior synthetic patient work. Z. Wang et al. (2024) demonstrate that expert-defined behavioural principles produce clinically plausible simulated patients. R. Wang et al. (2024) provide cognitive conceptualisation diagrams and conversational styles (direct, resistant, expressive, minimal, deviating, agreeable) that parallel the six archetypes used here. Six profiles sample the difficulty range but do not claim exhaustive coverage of patient variability (see section 7.2).

The generation stack traverses the full five-stage IRT protocol, producing complete therapy sessions that are then sliced to create evaluation entry points.

3.3. Frozen History Design

Frozen conversation histories serve as deterministic evaluation entry points. Each vignette's complete therapy session is sliced at rewriting-turn boundaries, where each slice ends after a complete therapist-patient exchange. Every trial, across all models and temperatures, receives the identical frozen history as input. This design eliminates context variation as a confound: if two trials produce different therapeutic responses, the difference is attributable to the model's stochastic sampling, not to differences in the input conversation.

A preliminary analysis tested Mistral Large 3 at five slice depths across all six vignettes at $T = 0.075$ and $T = 0.15$. Jaccard consistency was lowest at the second slice (mean $J = 0.77$ at $T = 0.075$) and rose to near-ceiling values from

the third slice onward ($J > 0.93$). The main experiment evaluates at the second rewriting-turn boundary (`slice_2`), making the measured stability a conservative estimate. Slice depth effects are reported in section 5.6.

3.4. Evaluation Stack

The evaluation stack bypasses the router and injects the rescripting prompt directly into the therapist model. Because the IRT stage is known a priori (all frozen histories are sliced within the rewriting phase), routing is unnecessary and would introduce an additional source of variation. Bypassing it isolates the therapist model from stage-classification error.

Each evaluation trial uses *fused generation*: the model produces a `<plan>` block declaring one to two therapeutic strategies, followed by the therapeutic response, in a single call. The plan tokens condition the response tokens in a Chain-of-Thought (CoT)-style generation, so the response is directly conditioned on the declared strategy. This fused approach enables the alignment analysis in Method 3 (subsection 3.7.3): because plan and response are generated in a single pass, any mismatch between declared intent and executed strategy reflects the model's own inconsistency rather than a decoupling artefact.

Twenty independent trials are collected per condition, yielding $\binom{20}{2} = 190$ pairwise comparisons. This trial count balances statistical power with compute cost: 190 pairs provide robust coverage of the pairwise similarity distribution while keeping the total experiment within budget.

3.5. The Plan Mechanism

The `<plan>` block raises a methodological question. CoT prompting (Wei et al., 2022) improves output structure by asking models to show their reasoning before answering. However, the faithfulness literature demonstrates that CoT explanations often do not reflect the model's actual computation. Turpin et al. (2023) show that biasing cues in the prompt alter model behaviour without appearing in the CoT trace. Lanham et al. (2023) find that early CoT tokens can be truncated or replaced without affecting the final answer, suggesting that the reasoning trace is reconstructed post-hoc rather than used for computation.

If a model declares its therapeutic strategy before responding, that declaration may be a post-hoc rationalisation rather than a window into its decision-making

process. This study resolves the tension by framing the `<plan>` block as *declared intent* rather than a reasoning trace. The plan captures what the model commits to, not what it internally computes. Whether or not the declaration reflects genuine reasoning, a practical consequence holds: if strategy declarations vary across runs, clinical responses will also vary. Variable declared intent predicts variable therapeutic behaviour, which is the clinically relevant property.

This framing sidesteps the faithfulness debate entirely while preserving three concrete uses. First, the declared strategies enable strategy-level consistency scoring (Method 1, subsection 3.7.1). Second, comparison between declared strategies and the executed response enables alignment verification (Method 3, subsection 3.7.3). Third, the plan provides a human-readable summary of the model's intended approach, supporting clinical oversight in deployment.

3.6. Strategy Taxonomy Development

A fixed taxonomy constrains the `<plan>` block to a predefined set of therapeutic strategies. This constraint is necessary for quantitative evaluation: free-text plans cannot be aggregated into set-similarity scores. The taxonomy evolved through four iterations to resolve a severe distribution skew that rendered early versions unusable.

The initial taxonomy (v1) defined eight categories including `empowerment` and `mastery`. Across 1,272 strategy picks, these two categories consumed 87.7% of all selections, while three categories (`social_support`, `sensory_modulation`, `gradual_exposure`) were never chosen. The root cause was a goal-mechanism conflation: both `empowerment` and `mastery` describe a therapeutic *goal* (the patient gains control) rather than a specific *mechanism* of change. Since every nightmare rescripting intervention serves the goal of increased mastery, the model defaulted to these labels regardless of the actual strategy employed.

Merging the two categories into a single `agency` category (v2) amplified the problem: the merged category achieved 100% dominance. A single super-category that applies to every rescripting scenario provides no discriminative signal.

The resolution in v3 was to split `agency` along the mechanism dimension. `Confrontation` describes external action toward the threat (the dreamer fights, challenges, or resists). `Self_empowerment` describes internal transformation within the dreamer (gaining abilities, growing stronger). These two categories are orthogonal: a dreamer who confronts a monster uses a different mechanism from one who discovers inner strength, even though both gain mastery in the outcome.

Table 3.1.: Strategy taxonomy (v4). Each category describes a rescripting mechanism rather than a therapeutic goal.

Category	Mechanism
`confrontation`	The dreamer directly faces, fights, or stands up to the threat
`self_empowerment`	The dreamer gains a new ability or inner strength
`safety`	A physical barrier, shelter, or protective figure shields the dreamer
`cognitive_reframe`	Existing dream elements change meaning
`social_support`	A new person, companion, or ally is introduced
`sensory_modulation`	Sensory qualities shift, including internal somatic calming

A final revision (v4) merged `emotional_regulation` into `sensory_modulation` after smoke testing showed that `emotional_regulation` appeared in only 1 of 120 trials (0.4%). In single-turn rescripting, internal calming manifests as sensory change (the dreamer feels calm, the environment becomes peaceful), making the boundary between the two categories clinically indistinct at this granularity.

The final six-category taxonomy captures concrete therapeutic mechanisms:

These categories are grounded in two clinical frameworks. Germain et al. (2004) identify six mastery dimensions in successful nightmare rescripting (physical, social, environmental, emotional, mystical, avoidance), all of which map to the taxonomy except avoidance. Harb et al. (2012) provide empirical rescripting frequencies from combat veterans, with all therapeutically productive categories covered. Avoidance and distancing are deliberately excluded: Harb et al. (2012) found that violence in rescripted dreams predicted worse treatment outcomes, and Krakow and Zadra (2006) frame IRT as "changing the nightmare" rather than escaping it. Reality markers (lucid dreaming cues) fall outside the scope of single-turn rescripting.

The iterative development process is documented here because taxonomy design directly shapes measured consistency. A coarse taxonomy inflates agreement scores (every trial picks the same super-category). A fine-grained taxonomy deflates them (minor strategy differences register as disagreements). The sensitivity of consistency metrics to category design is revisited in the discussion (section 6.5).

3.7. Evaluation Metrics

Three evaluation methods target distinct layers of the generation process. No single metric captures the full picture of clinical stability (section 2.4). Method 1 asks whether the model makes consistent therapeutic decisions. Method 2 asks whether it produces consistent therapeutic responses. Method 3 asks whether it does what it says it will do.

3.7.1. Method 1: Plan Consistency

Method 1 measures whether the model selects the same therapeutic strategies across independent trials. Strategy sets are extracted from the `<plan>` block of each trial output using the fixed taxonomy from Table 3.1. Pairwise Jaccard similarity quantifies agreement between strategy sets:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \tag{3.1}$$

The mean Jaccard score is computed over all $\binom{20}{2} = 190$ trial pairs per condition. Jaccard is appropriate because strategy combinations are unordered sets: a model that declares `confrontation / cognitive_reframe` has made the same clinical decision as one that declares `cognitive_reframe / confrontation`.

Because Jaccard operates on discrete sets of one to two items drawn from six categories, it takes only a small number of distinct values: $\{0, 0.33, 0.5, 0.67, 1.0\}$. This discreteness means Jaccard scores are ordinal rather than continuous, and descriptive statistics report medians and interquartile ranges rather than means and standard deviations.

Plan validity rate serves as an upstream quality gate. A malformed `<plan>` block that cannot be parsed into valid strategy identifiers is excluded from the Jaccard computation. The validity rate reports the proportion of trials with parseable plans, indicating whether the model can follow the structured output format at all.

3.7.2. Method 2: Response Similarity

Method 2 measures whether the model produces therapeutically equivalent responses across trials, independent of strategy declarations. Pairwise BERTScore F1 is computed over response texts (with plan blocks excluded) across the same 190 trial pairs.

The embedding model is DeBERTa-XLarge-MNLI (He et al., 2021). This model achieves the highest human correlation ($r = 0.778$) among more than 130 candidates on WMT16 segment-level evaluation. Its NLI fine-tuning aligns with the semantic equivalence question: determining whether two therapeutic responses convey the same clinical content, regardless of surface phrasing. Its disentangled attention mechanism handles the surface-form variation typical of therapeutic language, where a competent response naturally varies in wording while maintaining the same clinical intent.

Surface metrics (BLEU, ROUGE) are not used. As established in section 2.4, token-overlap metrics penalise valid therapeutic paraphrasing and cannot distinguish semantically equivalent from semantically divergent responses when surface forms differ.

3.7.3. Method 3: Plan-Response Alignment

Method 3 measures whether the model's therapeutic response implements its declared strategy. A separate LLM judge scores each declared strategy on a ternary scale:

- **0 (absent):** The strategy is not reflected in the response.
- **1 (partial):** The strategy is mentioned or briefly addressed but not fully developed.
- **2 (implemented):** The strategy is clearly and substantively present in the response.

This ternary scale is borrowed from the clinical fidelity literature (section 2.1). The ENACT (Kohrt et al., 2015) and NIH BCC (Bellg et al., 2004) frameworks use the same three-level gradation (absent, partial, implemented) to assess whether a human therapist delivers an intervention correctly. Applying the same scale to LLM outputs maintains continuity with established clinical evaluation practice.

The judge model is Gemini 3.1 Pro, accessed through Google AI Studio at $T = 1.0$. Gemini is from a different model family than all evaluation targets, preventing self-preference bias. Its thinking mode supports structured reasoning about whether a response implements a given strategy. The judge receives the strategy definitions, the declared strategies, and the response text, and produces a score with explicit justification for each strategy.

The judge runs at $T = 1.0$ rather than the conventional $T = 0.0$ because Gemini 3 models are architecturally optimised for their default temperature. Google's developer documentation warns that lowering the temperature below 1.0 can cause looping or degraded reasoning performance. Empirical evidence supports this choice: Renze (2024) finds no statistically significant impact of temperature changes between 0.0 and 1.0 on LLM problem-solving accuracy. Using Gemini 3.1 Pro as an LLM judge at its default temperature is consistent with recent practice: Liu and Tsai (2026)[1] employ it as an automated evaluator within a multi-agent assessment framework. Evaluation consistency is instead ensured through the structured ternary rubric and explicit scoring format.

Additional bias mitigations follow established recommendations (Zheng et al., 2023): cross-model judging ensures the judge never evaluates its own family, CoT justification requires the judge to explain its rating before assigning a score, and a transparent rubric anchors scores to specific behavioural criteria.

NLI cross-encoders were considered and rejected as an alternative. Testing showed an F1 ceiling of approximately 0.55 on the strategy-response alignment task, insufficient for reliable scoring. Cross-encoders also produce no reasoning trace, making it impossible to audit individual judgements.

Method 3 is labelled exploratory. Judge reliability is itself an open question: the judge's ternary scores are not a ground truth but a computational proxy for clinical fidelity assessment. Validation proceeds through human annotation of a random subset of approximately 50 trials, scoring the same rubric. Cohen's κ between human and judge provides a reliability estimate. If κ is low, Method 3 results carry the caveat explicitly.

Smoke testing showed alignment scores near ceiling (1.0) across all trials. If the full experiment confirms this pattern, alignment becomes a validation check (models consistently implement what they declare) rather than a comparative metric. This outcome would validate Method 1: if plans are faithfully executed, then Jaccard over plan sets measures genuine decision stability rather than label noise.

3.8. Model Selection

EU data sovereignty serves as the primary selection criterion for the study's main subject. The deployment context (reDreamAI) requires a model that can be self-

[1] Preprint, not yet peer-reviewed.

hosted within EU data residency boundaries (Dale, 2025). All evaluation targets run in non-thinking mode for fair comparison.

The **primary subject** is Mistral Large 3 (675B Mixture of Experts (MoE), 41B active parameters, Apache 2.0). It is the EU sovereign flagship and the model that reDreamAI would deploy in production. The research question centres on whether this model achieves stability comparable to proprietary alternatives.

Two additional EU-sovereign models complete the Mistral comparison. Mistral Small 4 (119B MoE, 6.5B active parameters, Apache 2.0) is Mistral's newest release and the most efficient MoE in the set, activating fewer parameters per token than any other model. Mistral Small 3.2 (24B dense, Apache 2.0) serves as a predecessor baseline, showing whether the architectural shift from dense to MoE affects stability.

The **proprietary ceiling** comprises two models. GPT-5.4 (non-thinking variant) is the strongest proprietary model without thinking overhead. OpenAI reintroduced user-configurable temperature with the non-reasoning variants of GPT-5.2, making controlled temperature comparison possible. Claude Sonnet 4.6 adds a second proprietary reference point; Anthropic's character training emphasises thoughtful, safety-aware interaction, which may influence therapeutic stability independently of raw capability.

Open-weight comparators span three size classes and two architectures:

The selection provides size-class diversity (24B–675B), architectural diversity (dense vs. MoE), and provider diversity across five vendors. Qwen 3.5 and Mistral Small 4 are hybrid-thinking models that support optional reasoning. To maintain fair comparison, reasoning is disabled for all evaluation targets.

DeepSeek V3.2 was initially included as a large MoE comparator but was replaced by Qwen 3.5 397B. DeepSeek applies internal temperature scaling (API $T = 1.0$ maps to internal $T \approx 0.3$), which would have complicated cross-model temperature comparisons. Qwen 3.5 397B occupies the same architectural niche without this complexity. Other models considered and dropped include Mistral Medium 3.1 (closed weights), GLM-5 (a third large MoE adds little signal), Nemotron 3 Super (requires $T = 1.0$ universally, complicating temperature comparison), and GPT-oss (mandatory reasoning mode cannot be disabled).

Table 3.2.: Evaluation targets. All models run in non-thinking mode via OpenRouter.

Group	Model	Size	Selection rationale
EU-sovereign (Mistral, Apache 2.0)			
Primary	Mistral Large 3	675B MoE (41B active)	EU-sovereign flagship. Production deployment candidate
EU	Mistral Small 4	119B MoE (6.5B active)	Newest Mistral. Fewest active parameters of any MoE in set
EU	Mistral Small 3.2	24B dense	Dense predecessor to Small 4
Open-weight comparators			
MoE	Qwen 3.5 122B	122B MoE (10B active)	Peer MoE comparator to Mistral Small 4
MoE	Qwen 3.5 397B	397B MoE (17B active)	Large MoE comparator to Mistral Large 3
Dense	Qwen 3.5 27B	27B dense	Same size class as Mistral Small 3.2
Dense	OLMo 3.1 32B	32B dense	Fully open (weights, data, code). Best provenance
Dense	Llama 3.3 70B	70B dense	Continuity with prior efficacy study
Proprietary ceiling			
Ceiling	Claude Sonnet 4.6	Proprietary	Character-trained for safety-aware interaction
Ceiling	GPT-5.4	Proprietary	Strongest non-thinking proprietary model

3.9. Experimental Conditions

The experiment follows a full factorial design:

Rather than testing only two temperature conditions, the experiment uses a five-point temperature sweep that spans from deterministic decoding to the vendor-recommended defaults. The scale is anchored at three points motivated by vendor documentation:

- $T = 0.0$ (greedy decoding): deterministic upper bound on stability. Any remaining inconsistency reflects floating-point non-determinism or infrastructure-level variation.
- $T = 0.15$: default temperature for Mistral models, taken from the HuggingFace generation configuration for Mistral Small 3.2.
- $T = 0.6$: default temperature for Qwen 3.5 and OLMo 3.1, also close to the standard Llama chat range (0.6–0.7) and the proprietary defaults (GPT-5.4 and Sonnet 4.6 default to $T = 0.7$).

The intermediate points $T = 0.075$ and $T = 0.3$ fill the gap between Mistral's low default and the Qwen/Llama range. $T = 0.075$ was added after an initial four-point run revealed that several models peak in consistency between $T = 0.0$ and $T = 0.15$, suggesting that a small amount of sampling variance can be beneficial (discussed in section 5.2).

This five-point sweep replaces a conventional binary comparison ($T = 0.0$ vs. a single stochastic point). It reveals the full degradation curve rather than a single gap, and allows each model to be evaluated at and around its vendor-recommended operating temperature.

Table 3.3.: Experimental conditions. Total: $6 \times 5 \times 20 = 600$ trials per model, 6,000 across all ten models.

Dimension	Values
Vignettes	anxious, avoidant, cooperative, resistant, skeptic, trauma
Trials	20 per condition
Temperatures	$T \in \{0.0,\ 0.075,\ 0.15,\ 0.3,\ 0.6\}$
Models	10 evaluation targets (Table 3.2)

3.10. Statistical Analysis

The analysis is primarily descriptive. For each model and temperature condition, central tendency and dispersion are reported for all three evaluation metrics, aggregated across vignettes. Jaccard scores are ordinal (discrete values from a small set), so they are summarised with medians and interquartile ranges. BERTScore F1 and alignment scores are reported with means and standard deviations.

Spearman rank correlations serve two purposes. First, per-model correlations between Jaccard and temperature quantify the temperature sensitivity of each model across the five-point sweep. Second, cross-method correlations (Method 1 vs. Method 2, Method 1 vs. Method 3, Method 2 vs. Method 3) assess whether the three evaluation layers capture redundant or independent information. If they diverge, the clinical implications differ: high plan consistency with low response similarity indicates stable strategy selections but variable execution, while the reverse indicates different strategies producing similar surface outputs. Spearman rather than Pearson is used because Jaccard scores are ordinal and BERTScore distributions may be non-normal.

Kruskal-Wallis tests assess whether models and vignettes differ significantly on each metric. These are exploratory rather than confirmatory: the study is designed to describe stability patterns, not to test a preregistered hypothesis.

No fixed threshold defines "stable enough." Clinical significance thresholds for LLM stability do not yet exist in the literature. Results are interpreted comparatively: model against model, across the temperature sweep. The random Jaccard baseline (expected value for random 2-of-6 selections, approximately 0.2) provides a lower reference point.

4. Implementation

This chapter describes how the methods from chapter 3 are realised in software. The pipeline is implemented in Python and structured around three concerns: a provider-agnostic LLM interface, a config-driven experiment runner, and a post-hoc aggregation pipeline. All source code is available in the project repository.

4.1. Software Architecture

The pipeline follows an async-first design. Every LLM call is non-blocking, and trials within a condition run concurrently via `asyncio.gather()`. This is a direct consequence of the factorial design: 360 trials per model would be prohibitive if executed sequentially. Parallel execution reduces wall-clock time from hours to minutes per model.

Configuration. Two YAML files define the full experiment declaratively. `models.yaml` specifies provider endpoints, model identifiers, role assignments, and evaluation targets. `experiment.yaml` controls sampling parameters (trial count, temperatures, maximum token length), metric selection, and output paths. Changing the evaluation target or temperature requires editing a single YAML field rather than modifying code. This separation ensures that experiment conditions are reproducible: the configuration snapshot saved with each run fully specifies how to recreate it.

Data validation. Pydantic models enforce type safety at every data boundary. The `Message` and `Conversation` classes validate all fields on construction and serialisation. When a frozen history is loaded from disk, `Conversation.from_dict()` uses `model_validate()` with `extra="ignore"`, silently discarding metadata fields that do not belong to the conversation schema. This is intentional: frozen histories may carry generation-time metadata (timestamps, model versions) that is irrelevant to evaluation.

The same validation applies to plan extraction. The `extract_plan_strategies()` function parses the `<plan>` block from each trial output and validates each strategy

Table 4.1.: Package structure. Each package maps to a distinct concern in the pipeline.

Package	Responsibility
`src/agents/`	Agent implementations (patient, therapist, router). Each agent extends a `BaseAgent` class that handles message formatting, system prompt injection, and provider delegation
`src/core/`	Data models (`Message`, `Conversation`), stage and language enums, YAML and prompt loaders
`src/llm/`	Provider abstraction. Single-class interface to all LLM backends
`src/stacks/`	Orchestration. `GenerationStack` runs the three-agent dialogue loop. `EvaluationStack` runs fused plan-and-response generation from frozen histories
`src/evaluation/`	Experiment runner, multi-trial sampler, and all three evaluation metrics (Jaccard, BERTScore, LLM judge alignment)

identifier against the taxonomy. A malformed plan that cannot be parsed into valid strategy identifiers fails at extraction time rather than propagating downstream where it would silently corrupt Jaccard scores.

Provider abstraction. A single `LLMProvider` class wraps an `AsyncOpenAI` client and exposes one async method: `generate(messages, **kwargs)`, returning a tuple of response content and token usage. All six evaluation targets, the patient agent, and the router use OpenRouter through this interface. The Gemini judge uses Google AI Studio through the same interface, since Google AI Studio exposes an OpenAI-compatible endpoint. Adding a new model or provider requires a configuration entry, not a code change.

A factory function (`create_provider()`) resolves role names (e. g. "therapist", "judge") to concrete provider configurations by looking up the active model assignment in `models.yaml`. For experiment-tier runs, the judge automatically upgrades from Gemini Flash Lite (used during development) to Gemini 3.1 Pro.

The codebase is organised into five packages:

4.2. Experiment Execution

An `ExperimentRun` object manages a single evaluation run: one model, one vignette, one temperature, twenty trials. On initialisation it creates a timestamped output directory and saves the configuration snapshot and frozen history.

Execution proceeds in three phases:

1. **Sampling.** A `Sampler` wraps the `EvaluationStack` and runs twenty independent trials. Each trial calls the therapist model once (fused mode), producing a `<plan>` block and a therapeutic response. Trials run in parallel via `asyncio.gather()` with built-in retry logic for rate-limit errors (HTTP 429) using exponential backoff.

2. **Metric computation.** After all trials complete, the run computes all three evaluation metrics. Method 1 (Jaccard) and Method 2 (BERTScore) are computed locally. Method 3 (alignment) issues parallel async calls to the Gemini judge, with a 120-second timeout per trial to handle unresponsive judge calls gracefully.

3. **Persistence.** Results are saved to the run directory.

Each run produces a self-contained output bundle:

```
experiments/runs/{timestamp}_{model}_{vignette}/
    config.yaml              # run parameters (model, temp, slice)
    frozen_history.json      # input conversation (irreplaceable)
    trials/
        trial_01.json        # plan + response + token usage
        ...
        trial_20.json
    metrics.json             # Jaccard, BERTScore, validity, alignment
    judgments.json            # raw judge outputs with reasoning
```

The frozen history is the only irreplaceable artefact in this structure. If the evaluation logic changes (a bug fix in Jaccard computation, a new judge prompt), metrics and judgements can be recomputed from the saved trials. Trials themselves can be recomputed from the frozen history plus the configuration, though at the cost of additional Application Programming Interface (API) calls.

The plan extraction logic handles several output formats that different models produce. The primary pattern matches a closed `<plan>` tag. Fallback patterns handle unclosed tags, bare tag-style strategy names, and plain-text first-line declarations. This robustness is necessary because the six evaluation targets do not all follow the structured output format identically, despite receiving the same system prompt.

4.3. Aggregation Pipeline

After all experiment runs complete, an aggregation script collects results into a single analysis frame. The script iterates over all run directories, loads each `config.yaml` and `metrics.json`, and produces one row per run indexed by model, vignette, slice, and temperature.

Strategy counts are expanded into six separate columns (one per taxonomy category), enabling per-strategy frequency analysis. Alignment scores per strategy are similarly expanded. The resulting DataFrame is saved as a CSV file (`stats/data/all_runs.csv`) that serves as the input for all downstream analysis and visualisation scripts.

This unified frame is a prerequisite for the cross-method analysis in section 5.5. Spearman correlations between Method 1, Method 2, and Method 3 require all three metrics aligned on the same condition index. Without a single frame that maps each condition to its Jaccard, BERTScore, and alignment scores, cross-method comparison would require manual alignment across separate output files.

Downstream scripts in `stats/scripts/` consume this CSV to produce descriptive statistics (`descriptives.py`), statistical tests (`tests.py`), and all five result figures. The separation between aggregation and analysis allows each step to be rerun independently: if a figure script is modified, only visualisation reruns, not the full aggregation.

5. Results

5.1. Plan Validity and Strategy Distribution

5.2. Plan Consistency (Method 1)

5.3. Response Similarity (Method 2)

5.4. Plan-Response Alignment (Method 3)

5.5. Cross-Method Analysis

5.6. Secondary Effects

6. Discussion

6.1. Interpreting the Three-Level Framework

6.2. Cross-Model Comparison

6.3. Temperature and Clinical Deployment

6.4. Vignette-Dependent Behaviour

6.5. Evaluation Framework Reflections

7. Conclusion

7.1. Summary of Findings

7.2. Limitations

7.3. Future Work

Acknowledgements

Use of AI Tools in This Thesis

Throughout the research and writing of this thesis, I made use of several LLM-based tools in a supporting capacity.

For intellectual engagement with the subject matter, I used Claude (Anthropic)[1] and ChatGPT (OpenAI)[2] as discussion partners. Their role was to challenge my reasoning, surface counterarguments, and suggest alternative framings for the arguments I had already developed. As English is not my native language, I occasionally used these tools to refine phrasing where my original formulations lacked precision or fluency.

For software development, I used Claude Code and Cursor (with Claude Sonnet) as coding assistants for the evaluation pipeline that forms the core of this thesis. Given that the pipeline itself measures LLM behavioral stability, I took particular care to verify all AI-generated code before inclusion, to avoid introducing confounds into the instrument being studied.

No part of the thesis text was generated by AI. All arguments, interpretations, and conclusions are my own.

[1] https://claude.ai
[2] https://chatgpt.com

Glossary

API Application Programming Interface

BDI Brief Dreamer Interview

CBT Cognitive Behavioral Therapy

CoT Chain-of-Thought

EU European Union

GDPR General Data Protection Regulation

IRT Imagery Rehearsal Therapy

JSON JavaScript Object Notation

LLM Large Language Model

MoE Mixture of Experts

NLI Natural Language Inference

NLP Natural Language Processing

PTSD Post-Traumatic Stress Disorder

RLHF Reinforcement Learning with Human Feedback

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A. Strategy Taxonomy

B. Prompts

C. Architecture Diagrams

D. Supplementary Results